

Outset · AI in Healthcare · Day 3 · Group 4

Fair for whom?

A skin-cancer screen you can actually audit -- because this dataset records skin tone -- and the dark-skin gap it exposes

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Outset

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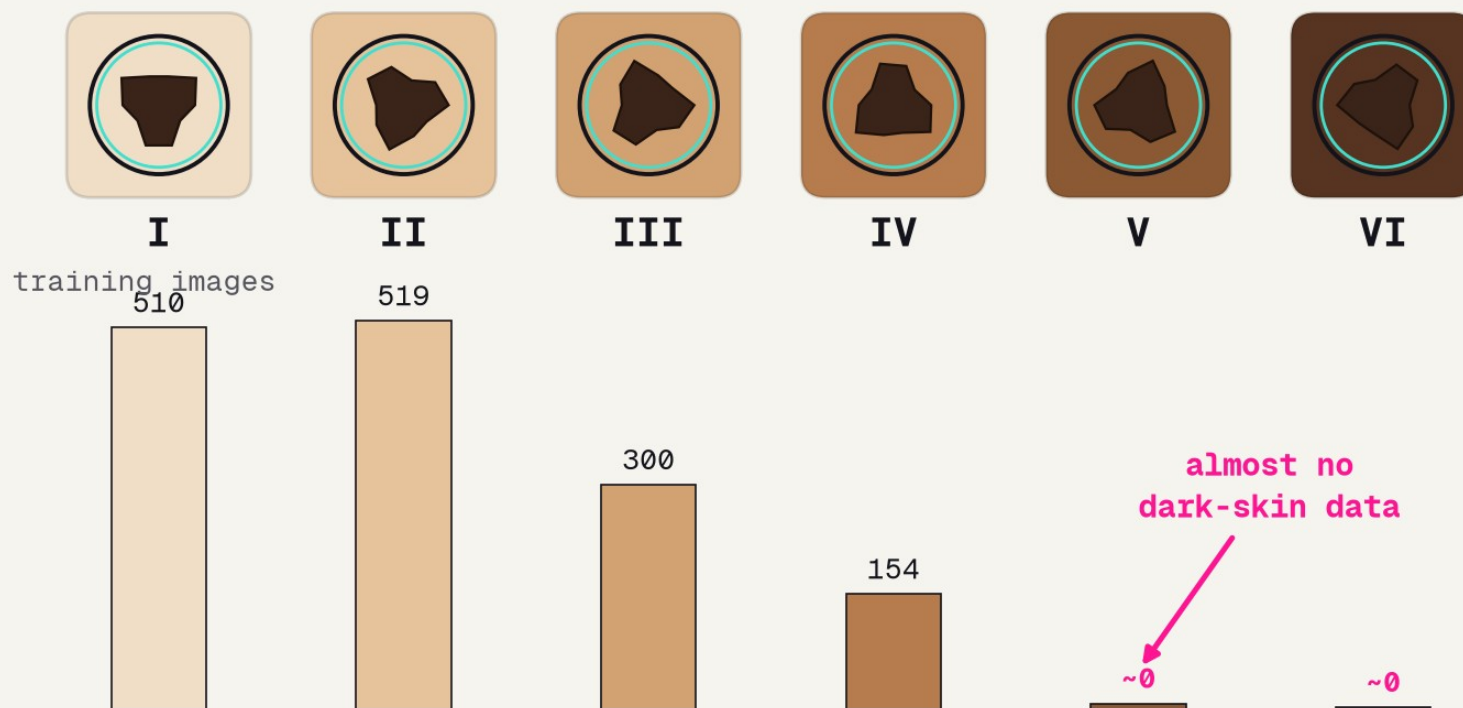
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BACKGROUND

The same lesion, six different skins

The same exam, far less practice on dark skin

one skin-cancer check, repeated across Fitzpatrick skin types I → VI



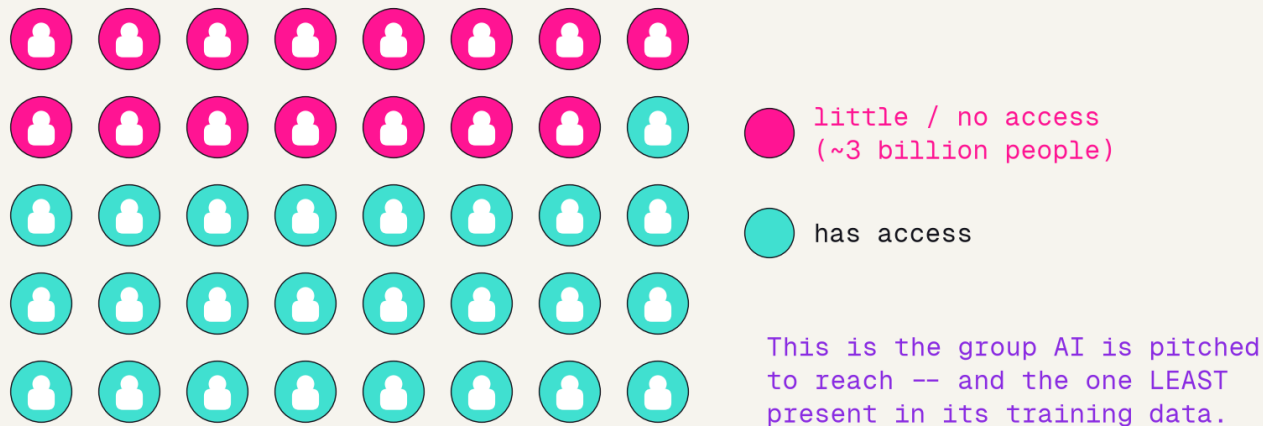
adapted from Wen et al. 2022 / Groh et al. 2021

The same skin-cancer check, from the lightest skin (I) to the darkest (VI) -- and how much training data any AI ever saw for each tone.

Who can even reach a dermatologist?

Roughly 3 billion people have little access to a dermatologist

each figure = about 200 million people; world population is around 8 billion



adapted from Buster et al. 2012 / Wen et al. 2022

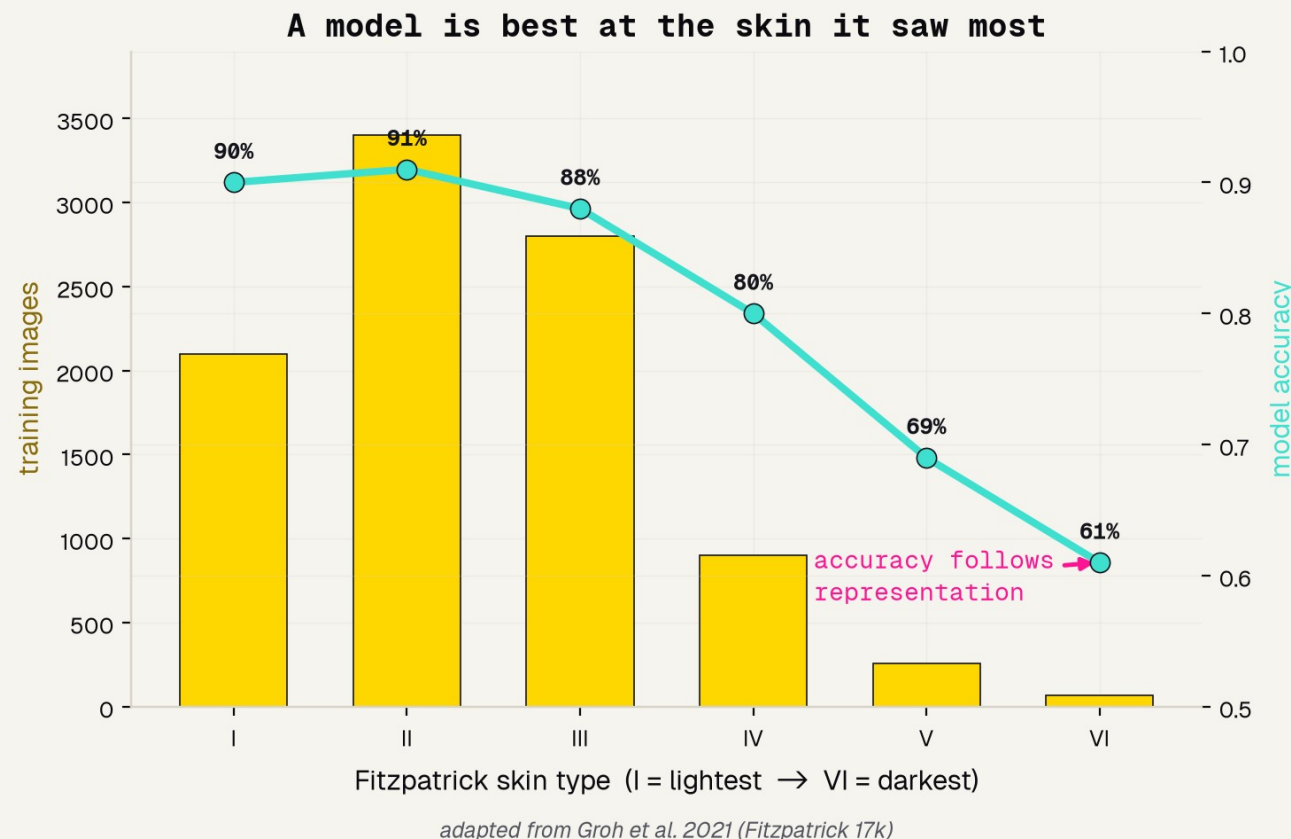
A PROMISE, AND AN IRONY

The promise: put a screen in a phone and reach the roughly three billion people who cannot see a dermatologist.

The irony: those same people -- often darker-skinned, and outside Europe and North America -- are the ones barely present in the training data.

A tool that works worst where care is scarcest can widen the very gap it was sold to close.

A model is best at the skin it saw most



16,577

clinical images Groh labeled by Fitzpatrick type

I → VI

the six skin types it audited separately

2021

Groh et al., CVPR -- the result we redraw

THE DATA

Why PAD-UFES-20

1,494

real clinical skin photos, each
biopsy-labeled malignant or
benign

skin type

PAD-UFES-20 records every
patient Fitzpatrick skin tone --
rare, and the whole reason we can
audit

**by
patient**

train and test never share a
patient, so the AUC we report is
honest

You can only audit what you record

MOST SKIN DATASETS

record the lesion -- and nothing about the patient. Skin tone is simply not in the file.



skin tone: unknown

CANNOT audit fairness by skin tone

PAD-UFES-20

records each patient Fitzpatrick skin type, so the score can be split apart and checked.



I-II

III-IV

V-VI

skin tone: recorded

CAN audit fairness by skin tone

THE MODEL

Borrow a brain, then split by patient

PRETRAINED

a network already trained on millions of everyday photos -- cats, cars, faces

it knows edges, colors, shapes



FINE-TUNE

train it further on PAD-UFES-20 clinical skin images

it learns malignant vs benign



SPLIT BY PATIENT

grade only on patients the model never saw during training

an honest AUC

The patient-level split is what keeps the score honest -- no person can appear in both training and test.

Model and data processing

1

MODEL

caformer_s18, pretrained, then fine-tuned with a fresh 2-class head

2

RESIZE

every clinical photo decoded to color and resized to 224x224 pixels

3

NORMALIZE

pixels rescaled to the fixed mean and scale the backbone was trained on

4

TASK

malignant (BCC, SCC, melanoma) vs benign -- one yes/no call

5

SPLIT & AUDIT

patient-level train/test split; then audit AUC by recorded skin tone

RESULTS

A working screen

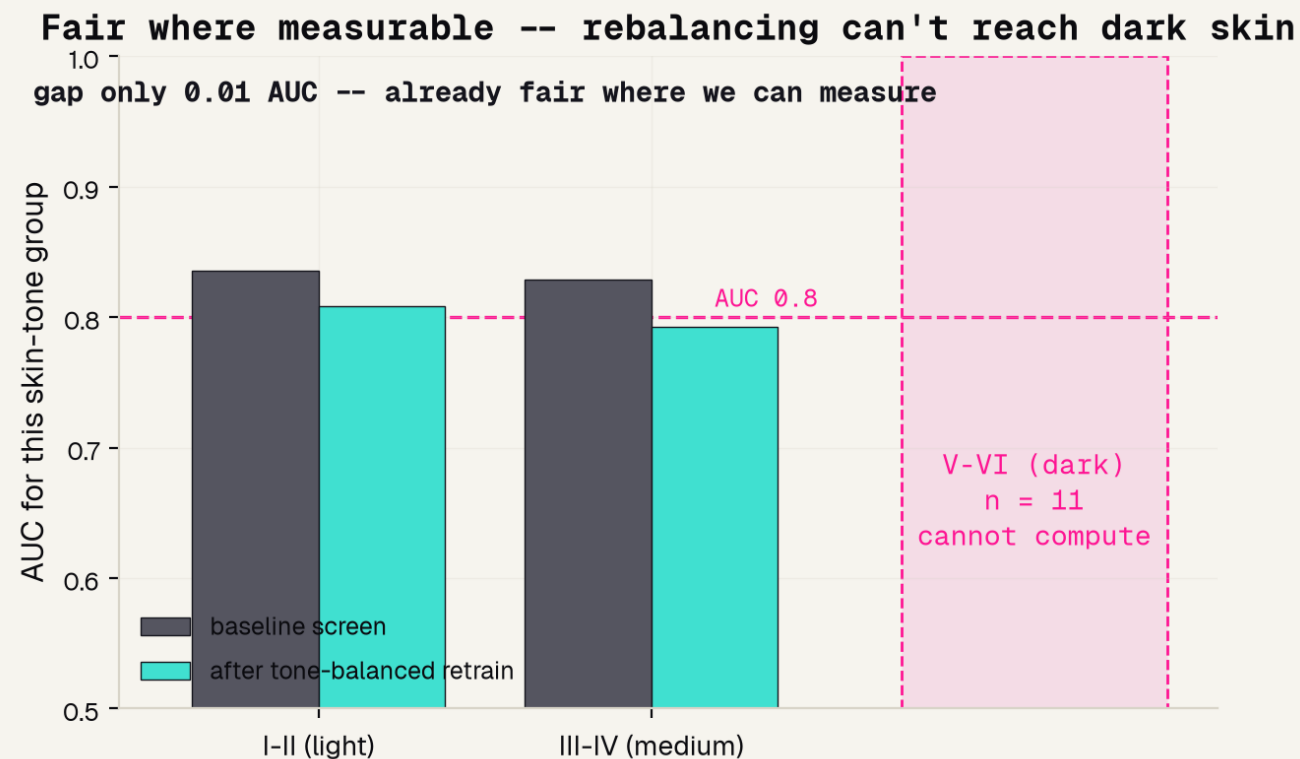
0.83

AUC on held-out patients

AUC is the area under the ROC curve: 0.5 is a coin flip, 1.0 is perfect. We grade on AUC, not accuracy, because a rare-cancer dataset rewards a lazy "always benign" guess.

A genuinely working screen -- which is exactly what makes the fairness audit worth doing.

Fair across the tones we can measure



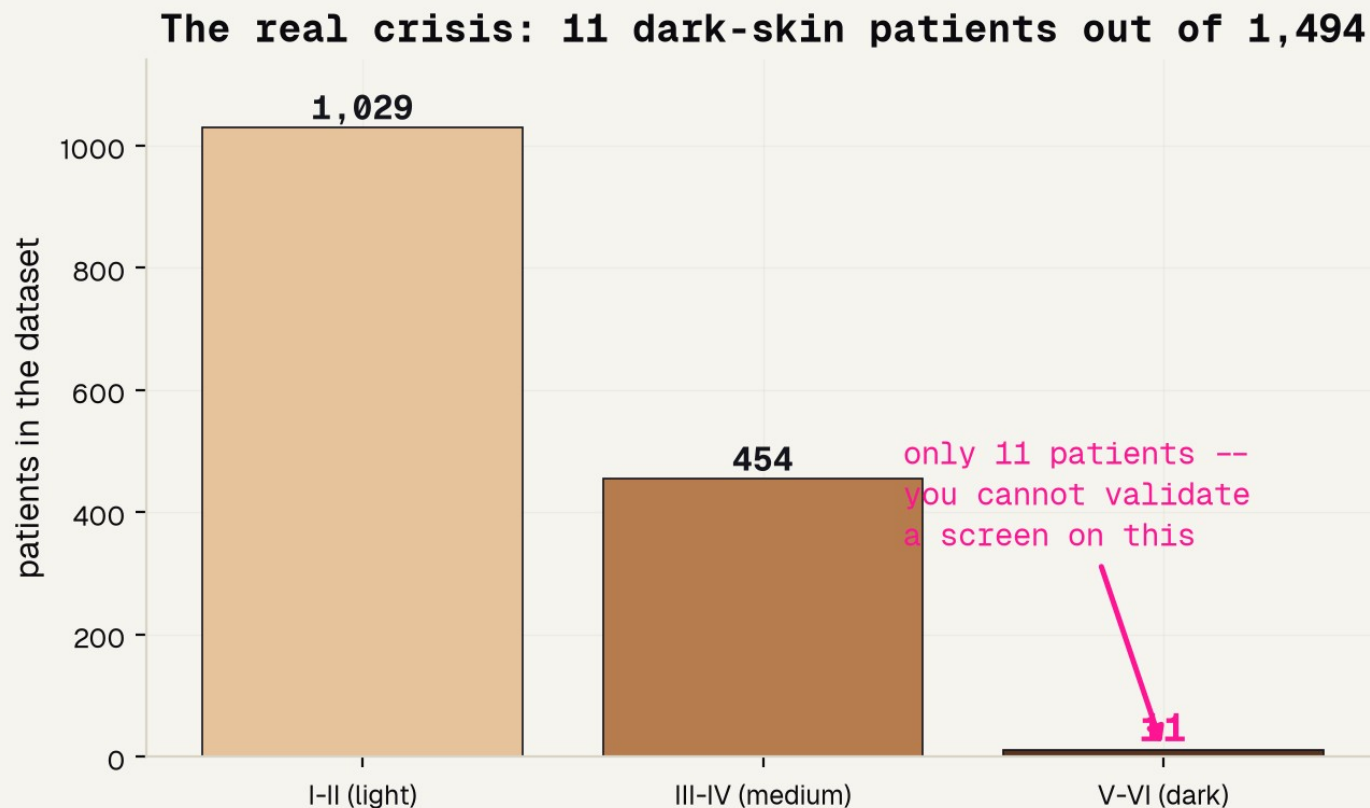
gap 0.01 AUC

light vs medium skin -- already roughly fair where we can actually measure it

dark skin: n = 11

a tone-balanced retrain cannot rebalance data that is not there -- the gap stays invisible

The real crisis: 11 patients



11

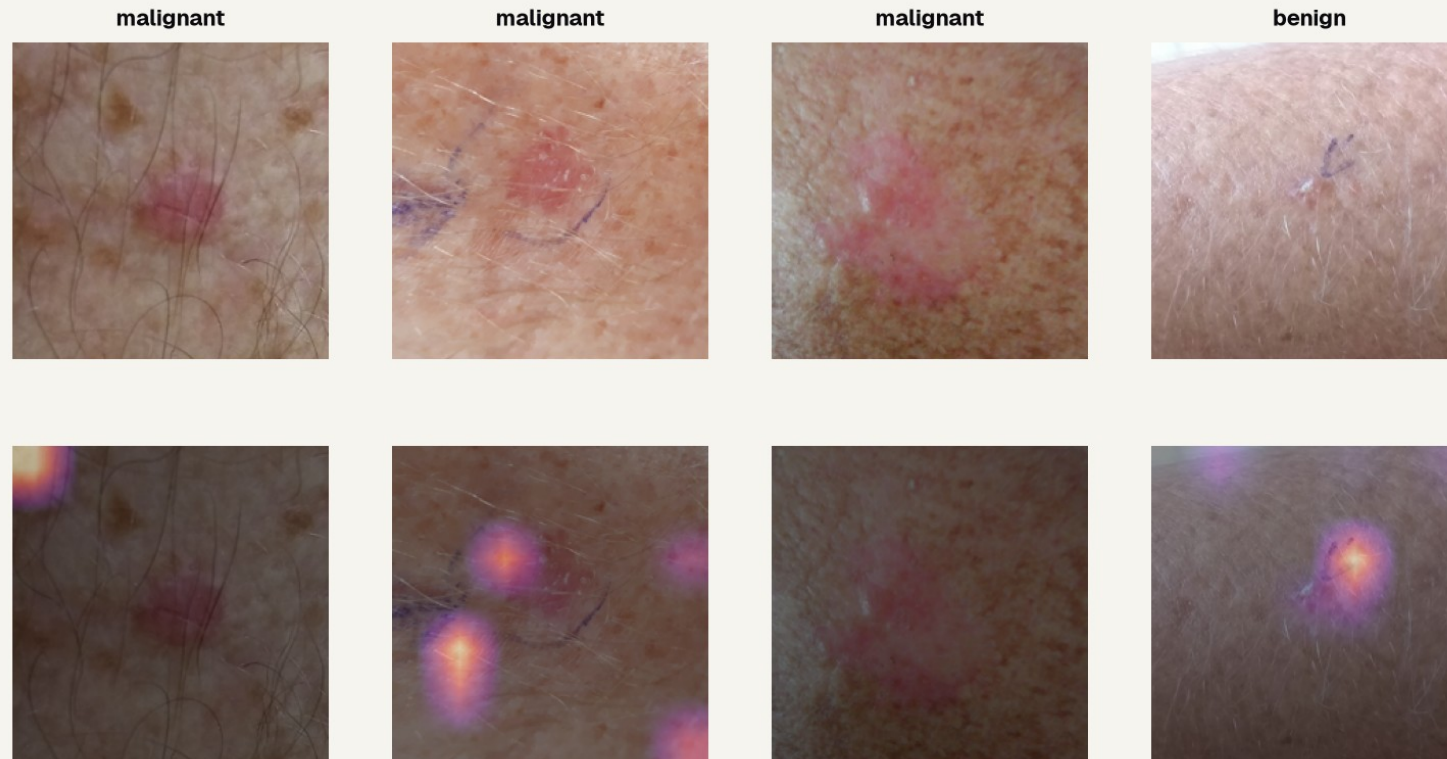
dark-skin patients, out of 1,494

You cannot validate a screen for a whole group of people from 11 of them -- no matter how good the overall average looks.

The under-representation is not a footnote. It is the finding.

Where the model looks

Where the model looks (after the fix)



Grad-CAM = feature importance for an image: the bright areas sit ON the lesion, the cues a clinician reads. It shows the model looks in the right PLACE -- not that it works on skin it never trained on.

BEING HONEST

What this project shows

IT WORKS, AND WE COULD AUDIT IT

AUC 0.83, and because PAD-UFES-20 records skin tone we could actually measure fairness -- which most skin datasets make impossible.

AUC 0.83 · auditable

THE GAP IS WHO IS MISSING

Only 11 dark-skin patients -- too few to validate. And no race or country is recorded, so those gaps stay invisible here.

11 dark-skin patients

THE FIX IS DATA, NOT A TRICK

Rebalancing cannot invent dark-skin patients. Fixing dark-skin performance means collecting representative data -- DDI, Fitzpatrick 17k.

collect, do not resample

References

GAPS & LANDMARKS

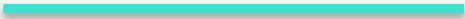
- [1] Fitzpatrick 1988, Arch Dermatol -- the six-point skin-type scale.
- [2] Groh et al. 2021, CVPR Workshops -- Fitzpatrick 17k; accuracy tracks representation.
- [3] Daneshjou et al. 2022, Sci Adv -- DDI; top models drop on dark skin.
- [4] Obermeyer et al. 2019, Science -- a fair-looking algorithm that underserved Black patients.

DATA & ACCESS

- [5] Wen et al. 2022, Lancet Digital Health -- most public skin images are light skin.
- [6] Adamson and Smith 2018, JAMA Dermatology -- AI can widen dermatology care gaps.
- [7] Buster, Stevens and Elmetts 2012, Dermatologic Clinics -- disparities and access.

The honest bottom line

A model can only be fair to the people the data remembers.



This screen works, and we could audit it only because the data recorded skin tone.
Record who the patient is -- then go collect the data the model is missing.

Thank You

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夢想